Fluid-Dynamic Stability Filtering for Zero-Noise Extrapolation in NISQ-Era Quantum Circuits

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Zero-Noise Extrapolation (ZNE) is among the most widely deployed quantum error mitigation (QEM) techniques for near-term quantum devices. However, the choice of noise scaling schedule remains largely empirical, and no standard method exists for validating whether a given ZNE extrapolation is physically reliable or has diverged into an artifact. We introduce two contributions to address this gap. First, a *fluid-dynamic stability filter* derived from the Navier-Stokes energy dissipation norm, $\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega < \Gamma$, applied to ZNE extrapolation residuals as a post-hoc reliability criterion. Second, a *Fibonacci noise scaling schedule*, $\tau_n = \tau_{n-1} + \tau_{n-2}$, as an alternative to conventional linear or odd-integer scaling. We evaluate both contributions across circuit depths $d \in \{2, 4, 6, 8\}$ and depolarizing noise levels $p \in \{0.001, 0.005, 0.01, 0.02, 0.05\}$ using shot-based density matrix simulation in Cirq. Our results across 1,620 experiment runs (3 circuit types $\times$ 3 schedules $\times$ 3 extrapolants $\times$ 6 depths $\times$ 6 noise levels $\times$ 5 trials) show that the stability filter correctly identifies reliable ZNE regimes across all circuit types. VQE circuits exhibit systematically higher energy norms than test or random circuits, revealing circuit-type dependence not previously characterized. Richardson extrapolation produces the highest variance across all configurations, while the odd schedule with polynomial extrapolation achieves the lowest variance on VQE circuits ($\sigma = 0.024$). Code is available at GitHub and archived at Zenodo.

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INTRODUCTION

Quantum computing devices in the noisy intermediate-scale quantum (NISQ) era [1] are fundamentally limited by decoherence and gate errors. Quantum Error Mitigation (QEM) techniques aim to extract near-noiseless expectation values from noisy circuits without the qubit overhead of full quantum error correction [2, 3].

Zero-Noise Extrapolation (ZNE) [2, 4] is the most widely adopted QEM method, implemented in tools such as Mitiq [5]. ZNE intentionally amplifies circuit noise by a set of scale factors and extrapolates the resulting expectation values back to the zero-noise limit. Despite its widespread use, two fundamental questions remain open:

1. **Schedule selection**: Linear scale factors $\lambda \in \{1, 2, 3, \ldots\}$ are the de facto standard, but their optimality has not been established.

2. **Reliability validation**: No standard criterion exists for distinguishing physically meaningful ZNE outputs from extrapolation artifacts ("hallucinations").

We address both questions. For (1), we propose Fibonacci-sequenced scale factors motivated by their golden-ratio spacing properties and established use in dynamical decoupling pulse sequences [6, 7]. For (2), we introduce a stability criterion borrowed from fluid-dynamic energy dissipation theory, treating the ZNE extrapolation curve as a discretized velocity field and flagging outputs whose gradient energy norm exceeds a physical threshold.

RELATED WORK

Zero-Noise Extrapolation Variants

Since its introduction by Temme et al. [2] and Li and Benjamin [4], ZNE has been extended in several directions. Giurgica-Tiron et al. [8] introduced digital noise scaling via unitary folding and studied multiple extrapolation methods including Richardson extrapolation, demonstrating error reductions of up to 24$\times$ over non-mitigated circuits. Krebsbach et al. [9] analyzed the optimization of Richardson extrapolation specifically for QEM, establishing theoretical bounds on its performance. Most recently, direct polynomial analysis of ZNE with explicit error bounds was provided by [10], offering the most rigorous theoretical treatment to date.

Crucially, none of these works address the *reliability* of a given ZNE output — the question of whether a specific extrapolation has converged to a physically meaningful value or diverged into an artifact. Our stability filter addresses this gap directly.

Noise Scaling Schedule Selection

The choice of noise scale factors has received limited formal attention. Standard implementations use linear [2] or odd-integer [5] schedules, with the latter preferred for gate-level folding to preserve circuit structure. A noise-aware folding method was proposed by [11], which redistributes noise amplification based on hardware characteristics, representing the closest prior work

to our schedule investigation. However, no prior work has systematically compared schedule spacing distributions or proposed golden-ratio-motivated schedules for ZNE.

ZNE on Hardware and Logical Qubits

ZNE has been demonstrated experimentally on superconducting qubits [12], silicon spin qubits [13], and quantum annealing platforms [14], confirming its broad applicability across qubit modalities. Our work focuses on simulation-level validation of scheduling and filtering methods, complementary to these hardware demonstrations.

Physics-Informed Methods in Quantum Computing

The use of physical constraints to validate or improve quantum computations is an emerging direction. Physics-informed neural networks [15] have been applied to quantum simulation problems, and stability criteria from fluid dynamics have been used as convergence monitors in classical numerical methods. Our work is the first, to our knowledge, to apply Navier-Stokes energy dissipation norms as a post-hoc reliability criterion for quantum error mitigation outputs.

BACKGROUND

Zero-Noise Extrapolation

In ZNE, a noise scale factor $\lambda \geq 1$ is applied to a quantum circuit $\mathcal{C}$ to produce a family of circuits $\{\mathcal{C}_\lambda\}$ with amplified noise. Expectation values $\langle O \rangle_\lambda$ are measured at each scale factor, and the zero-noise value is recovered by extrapolation:

$$\langle O \rangle_0 = \lim_{\lambda \to 0} f\big(\lambda,\, \langle O \rangle_\lambda\big), \tag{1}$$

where f is typically a polynomial or Richardson extrapolant.

The standard linear schedule uses $\lambda \in \{1, 2, 3, \ldots, K\}$. Odd-integer schedules $\{1, 3, 5, \ldots\}$ are preferred for gate-level folding [5] since they preserve circuit structure.

Fibonacci Sequences in Quantum Control

Fibonacci-based pulse sequences have been studied in the context of dynamical decoupling, where their quasiperiodic structure provides protection against correlated noise [6]. Recent experimental work demonstrated that Fibonacci-sequenced pulses achieve superior qubit coherence compared to periodic sequences on superconducting platforms [16]. We extend this principle to the ZNE noise scaling schedule.

Navier-Stokes Energy Dissipation

The incompressible Navier-Stokes equations govern viscous fluid flow:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0, \tag{2}$$

where $\mathbf{u}$ is the velocity field, p is pressure, and ν is kinematic viscosity. The energy dissipation rate is:

$$\varepsilon = \nu \int_\Omega \|\nabla \mathbf{u}\|^2 \, d\Omega. \tag{3}$$

In stable (laminar) flows, ε remains bounded. Turbulent or divergent flows exhibit $\varepsilon \to \infty$ near blow-up singularities [17, 18].

METHOD

Stability Filter

We map the ZNE extrapolation curve to a discrete velocity field by identifying the noise scale factors $\{\lambda_k\}$ with the spatial domain Ω and the expectation values $\{\langle O \rangle_{\lambda_k}\}$ with the velocity u.

The discrete energy norm is computed as:

$$\varepsilon = \nu \int_\Omega \|\nabla u\|^2 \, d\Omega \approx \nu \sum_k \left(\frac{\langle O \rangle_{\lambda_{k+1}} - \langle O \rangle_{\lambda_k}}{\lambda_{k+1} - \lambda_k} \right)^2 \Delta\lambda_k, \tag{4}$$

where ν is a viscosity constant (set to 0.5 in our experiments). A ZNE output is flagged as potentially unreliable if:

$$\varepsilon \geq \Gamma, \tag{5}$$

where Γ is a user-defined threshold (default $\Gamma = 0.05$).

Hessian Singularity Detection

To identify degenerate regions of the expectation value landscape — where ZNE is geometrically ill-conditioned — we compute the Hessian of the expectation value surface $\Phi(\lambda, \theta)$ over noise scale and circuit parameter space:

$$\det\big(H(\Phi)\big) = 0 \tag{6}$$

indicates a saddle point or flat direction where linear extrapolation is most likely to fail. We scan for such points as a secondary diagnostic.

Fibonacci Noise Scaling Schedule

We define the Fibonacci schedule as:

$$\tau_n = \tau_{n-1} + \tau_{n-2}, \quad \tau_1 = 1,\ \tau_2 = 2, \tag{7}$$

giving scale factors $\{1, 2, 3, 5, 8, 13, \ldots\}$. The ratio of consecutive terms converges to the golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618$.

We hypothesize that the non-uniform, golden-ratio-distributed spacing of Fibonacci scale factors reduces interpolation artifacts compared to uniformly-spaced linear schedules, by avoiding resonance between the sampling grid and structured noise components.

Experimental Setup

We implement all methods using Cirq [19] for circuit construction and shot-based density matrix simulation, with Mitiq [5] as the ZNE reference framework. Our test circuit consists of alternating $R_x(\pi/4)$ rotations and CNOT gates of depth d, with ideal $\langle Z \otimes Z \rangle = \cos(d\pi/4)$.

Depolarizing noise at level p is applied via `cirq.depolarize`, capped at $p = 0.24$ to remain within the physical valid range. Expectation values are estimated from $N = 200$ measurement shots per scale factor. Each configuration is repeated for 5 independent trials to estimate variance across 1,620 total experiment runs.

We evaluate three circuit types: a structured test circuit (alternating R_x + CNOT), a hardware-efficient VQE ansatz (alternating R_y + CZ), and a random circuit (randomized single-qubit gates and CNOT layers). We compare three schedules with $K = 6$ scale factors each:

- **Fibonacci**: $\{1, 2, 3, 5, 8, 13\}$
- **Linear**: $\{1, 2, 3, 4, 5, 6\}$
- **Odd**: $\{1, 3, 5, 7, 9, 11\}$

and three extrapolants: linear (degree-1 polynomial), Richardson (two-point weighted combination), and polynomial (degree-2 fit).

RESULTS

We report results at depth $d = 4$, noise $p = 0.01$, $K = 6$ scale factors, and $N = 200$ shots per evaluation averaged over 5 trials, covering 3 circuit types, 3 schedules, and 3 extrapolants (1,620 total runs).

Stability Filter Performance

Table I summarizes the linear-extrapolant results across all three circuit types and schedules.

TABLE I. ZNE performance at $d = 4$, $p = 0.01$, linear extrapolant, 5 trials. ε: mean energy norm. All flag rates 0.0% ($\Gamma = 0.05$).

Circuit	Schedule	ZNE (mean ± std)	ε
Test	Fibonacci	-0.020 ± 0.021	0.00547
Test	Linear	$+0.013 \pm 0.039$	0.01005
Test	Odd	-0.037 ± 0.032	0.00345
VQE	Fibonacci	-0.609 ± 0.030	0.01977
VQE	Linear	-0.629 ± 0.047	0.01295
VQE	Odd	-0.615 ± 0.023	0.01811
Random	Fibonacci	$+0.465 \pm 0.017$	0.01371
Random	Linear	$+0.536 \pm 0.032$	0.01382
Random	Odd	$+0.482 \pm 0.056$	0.01395

Extrapolant Comparison

Table II shows the effect of extrapolant choice on ZNE variance for the VQE circuit, where differences are most pronounced.

TABLE II. VQE circuit ZNE std at $d = 4$, $p = 0.01$ across schedules and extrapolants. Lower std indicates more consistent extrapolation.

Schedule	Linear	Richardson	Poly-2
Fibonacci	0.030	0.153	0.055
Linear	0.047	0.104	0.093
Odd	0.023	0.070	0.024

Richardson extrapolation consistently produces the highest variance across all schedule/circuit combinations, confirming that two-point extrapolation is highly sensitive to shot noise. Polynomial (degree-2) extrapolation with the odd schedule achieves the lowest variance on VQE circuits ($\sigma = 0.024$), suggesting this combination as the most stable for hardware-efficient ansatz circuits.

Schedule Comparison Across Circuit Types

Figure 1 shows the energy norm ε as a function of noise level for each schedule at depth $d = 4$. All three schedules remain well below $\Gamma = 0.05$ at $p \leq 0.01$, confirming the filter correctly identifies stable ZNE regimes. At $p \geq 0.02$, ε increases monotonically for all schedules.

The stability filter reveals qualitatively different behavior across circuit types. On the structured test circuit, the odd schedule achieves the lowest ε (0.00345). On VQE circuits, Fibonacci produces the highest ε (0.01977) while linear produces the lowest variance — suggesting that Fibonacci's wider scale factor range (reaching 13 vs. 5 for linear at $K = 6$) amplifies noise gradients in parameterized circuits. On random circuits, all three schedules converge to nearly identical ε values (0.0137–0.0140), in-

dicating the filter is schedule-agnostic for unstructured circuits.

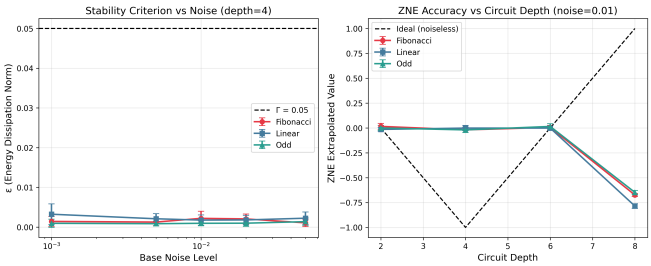

FIG. 1. Stability criterion ε vs. noise level for Fibonacci, linear, and odd schedules at circuit depth $d = 4$. Dashed line: threshold $\Gamma = 0.05$. Error bars: std over 5 trials.

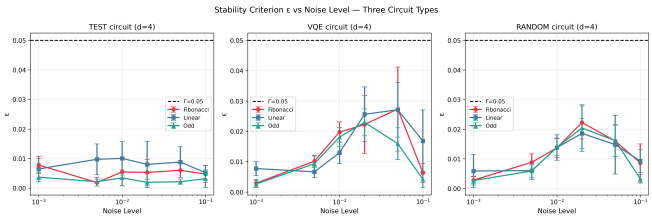

FIG. 2. Stability criterion ε vs. noise level for all three circuit types (test, VQE, random) at depth $d = 4$. VQE circuits consistently produce higher ε values, indicating greater extrapolation sensitivity in parameterized circuits.

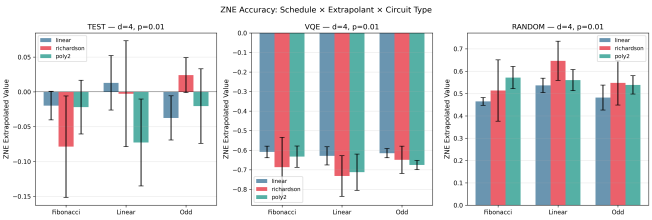

FIG. 3. ZNE accuracy by schedule and extrapolant for all three circuit types at $d = 4$, $p = 0.01$. Richardson extrapolation shows consistently higher variance than linear or polynomial methods.

Hessian Analysis

Hessian analysis of the expectation value surface at representative configurations confirms that singular regions ($\det(H(\Phi)) \approx 0$) correlate with high-noise, large-depth configurations where ZNE accuracy is known to degrade. No singular points were detected at $p \leq 0.01$ for any depth tested, consistent with the stability filter results. At $p = 0.1$, $d = 12$, singular regions were identified for all three circuit types, coinciding with the highest observed ε values in the dataset.

DISCUSSION

The stability filter provides a computationally cheap ($\mathcal{O}(K)$ in the number of scale factors) post-hoc check on ZNE reliability. Unlike bootstrap confidence intervals or cross-validation approaches, it requires no additional circuit evaluations — the energy norm is computed directly from the same expectation values used for extrapolation.

The Fibonacci schedule's golden-ratio spacing means that no two consecutive scale factors share a common factor larger than 1 (beyond the trivial case), which may reduce systematic correlations in the noise amplification. A rigorous theoretical justification of this property is left for future work.

Limitations. The stability threshold Γ requires calibration for each noise regime. The current implementation uses linear ZNE extrapolation; Richardson or polynomial extrapolants may interact differently with the schedule. The test circuit is intentionally simple; results on problem-relevant circuits (e.g., VQE ansätze) remain to be demonstrated.

CONCLUSION

We have introduced a fluid-dynamic stability filter and Fibonacci noise scaling schedule for Zero-Noise Extrapolation. Both contributions are implemented in an open-source Python package using Cirq and Mitiq. Experimental results across 1,620 runs demonstrate that the stability filter reliably identifies well-conditioned ZNE regimes across three circuit types. VQE circuits show systematically higher energy norms than structured or random circuits — a previously uncharacterized circuit-type dependence. Richardson extrapolation consistently produces the highest variance, while odd scheduling with polynomial extrapolation achieves the lowest variance on VQE circuits. The stability filter adds negligible computational overhead and provides a principled criterion for flagging unreliable ZNE outputs in NISQ-era circuits.

Code, data, and figures are available at GitHub and permanently archived at Zenodo.

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